A Living Review of Quantum Information Science in High Energy Physics

Pamela Pajarillo, Sokratis Trifinopoulos, Jesse Thaler

Center for Theoretical Physics, Massachusetts Institute of Technology
Cambridge, MA 02139, USA

E-mail: pampaja@mit.edu, trifinos@mit.edu, jthaler@mit.edu

ABSTRACT: Inspired by "A Living Review of Machine Learning for Particle Physics"[1], the goal of this document is to provide a nearly comprehensive list of citations for those developing and applying quantum information approaches to experimental, phenomenological, or theoretical analyses. Applications of quantum information science to high energy physics is a relatively new field of research. As a living document, it will be updated as often as possible with the relevant literature with the latest developments. Suggestions are most welcome.

[1] See https://github.com/iml-wg/HEPML-LivingReview.

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1 Introduction

The purpose of this note is to collect references for quantum information science as applied to particle and nuclear physics. The papers are listed in reverse chronological order. In order to be as useful as possible, this document will continually change. Please check back [2] regularly. You can simply download the .bib file to get all of the latest references. Suggestions are most welcome.

2 Nuclear and Particle Physics in Quantum Information Science

2.1 Reviews and Whitepapers

2.1.1 Reviews

2.1.1.1 Quantum entanglement and Bell inequality violation at colliders [1]

[2] See https://github.com/PamelaPajarillo/NUPAQIS-LivingReview.

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

2.1.1.2 Quantum Computing Applications in Future Colliders [2]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.1.1.3 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 2021

2.1.2 Whitepapers

2.1.2.1 Quantum Computing for High-Energy Physics: State of the Art and Challenges [4]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.1.2.2 Report of the Snowmass 2021 Theory Frontier Topical Group on Quantum Information Science [5]

- **Authors:** Simon Catterall, Roni Harnik, Veronika E. Hubeny, Christian W. Bauer, Asher Berlin, Zohreh Davoudi, Thomas Faulkner, Thomas Hartman, Matthew Headrick, Yonatan F. Kahn, Henry Lamm, Yannick Meurice, Surjeet Rajendran, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 29 September 2022

2.1.2.3 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [6]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

2.1.2.4 Quantum computing hardware for HEP algorithms and sensing [7]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Cancelo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti
- **Posted on arXiv:** 18 April 2022

2.1.2.5 Quantum Simulation for High-Energy Physics [8]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.1.2.6 Quantum Networks for High Energy Physics [9]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie
- **Posted on arXiv:** 31 March 2022

2.1.2.7 New Horizons: Scalar and Vector Ultralight Dark Matter [10]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou
- **Posted on arXiv:** 28 March 2022

2.1.2.8 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.1.2.9 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [12]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

2.1.2.10 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [13]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

2.1.2.11 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [14]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

2.1.3 Quantum Simulations

2.1.3.1 Review on Quantum Computing for Lattice Field Theory [15]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in PoS:** 01 February 2023

2.2 Anomaly Detection

2.2.1 Continuous Variable Quantum Computing

2.2.1.1 Unsupervised event classification with graphs on classical and photonic quantum computers [16]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

2.2.2 Quantum Annealing

2.2.2.1 A Quantum Algorithm for Model-Independent Searches for New Physics [17]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

2.2.3 Quantum Autoencoders

2.2.3.1 Anomaly detection in high-energy physics using a quantum autoencoder [18]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi
- **Posted on arXiv:** 09 December 2021
- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

2.2.4 Quantum Kernel Methods

2.2.4.1 Quantum anomaly detection in the latent space of proton collision events at the LHC [19]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

2.2.4.2 Unravelling physics beyond the standard model with classical and quantum anomaly detection [20]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

2.2.5 Quantum Unsupervised Clustering Algorithms

2.2.5.1 Quantum anomaly detection in the latent space of proton collision events at the LHC [19]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

2.2.5.2 Unsupervised event classification with graphs on classical and photonic quantum computers [16]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

2.2.6 Variational Quantum Circuits

2.2.6.1 Quantum anomaly detection for collider physics [21]

- **Authors:** Sulaiman Alvi, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 16 June 2022
- **Published in JHEP:** 22 February 2023

2.2.6.2 Anomaly detection in high-energy physics using a quantum autoencoder [18]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi
- **Posted on arXiv:** 09 December 2021
- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

2.3 Beyond the Standard Model

2.3.1 Quantum Algorithms Based on Amplitude Amplification

2.3.1.1 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [22]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

2.3.1.2 Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider [23]

- **Authors:** Anthony E. Armenakas, Oliver K. Baker
- **Posted on arXiv:** 01 October 2020

2.3.2 Quantum Annealing

2.3.2.1 Completely quantum neural networks [24]

- **Authors:** Steve Abel, Juan C. Criado, Michael Spannowsky
- **Posted on arXiv:** 23 February 2022
- **Published in Phys.Rev.A:** 01 August 2022

2.3.2.2 Quantum algorithm for the classification of supersymmetric top quark events [25]

- **Authors:** Pedrame Bargassa, Timothèe Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothèe Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

2.3.3 Quantum Kernel Methods

2.3.3.1 Unravelling physics beyond the standard model with classical and quantum anomaly detection [20]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

2.3.4 Quantum Sensors

2.3.4.1 Searching for Dark Matter with a Superconducting Qubit [26]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

2.3.5 Variational Quantum Circuits

2.3.5.1 Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets [27]

- **Authors:** Miguel Caçador Peixoto, Nuno Filipe Castro, Miguel Crispim Romão, Maria Gabriela Jordão Oliveira, Inês Ochoa
- **Posted on arXiv:** 06 November 2022
- **Published in Front. Artif. Intell. 6 (2023) 1268852:** 20 November 2023

2.4 Cosmology and Early Universe Physics

2.4.1 Quantum Annealing

2.4.1.1 Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer [28]

- **Authors:** João Caldeira, Joshua Job, Steven H. Adachi, Brian Nord, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019

2.5 Detector Simulation

2.5.1 Continuous Variable Quantum Computing

2.5.1.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [29]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

2.5.2 Quantum Annealing

2.5.2.1 Hybrid actor-critic algorithm for quantum reinforcement learning at CERN beam lines [30]

- **Authors:** Michael Schenk, Elías F. Combarro, Michele Grossi, Verena Kain, Kevin Shing Bruce Li, Mircea-Marian Popa, Sofia Vallecorsa
- **Posted on arXiv:** 22 September 2022
- **Published in Quantum Sci.Technol.:** 21 February 2024

2.5.3 Quantum Generative Adversarial Networks

2.5.3.1 Running the Dual-PQC GAN on noisy simulators and real quantum hardware [31]

- **Authors:** Su Yeon Chang, Edwin Agnew, Elías F. Combarro, Michele Grossi, Steven Herbert, Sofia Vallecorsa
- **Posted on arXiv:** 30 May 2022
- **Published in J.Phys.Conf.Ser.:** 2023

2.5.3.2 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [32]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

2.5.3.3 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [29]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

2.5.4 Variational Quantum Circuits

2.5.4.1 Precise image generation on current noisy quantum computing devices [33]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borras, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

2.6 Elementary Particle Systems

2.6.1 Quantum Information Theory

2.6.1.1 Full quantum tomography of top quark decays [34]

- **Authors:** J.A. Aguilar-Saavedra
- **Posted on arXiv:** 22 February 2024
- **Published in Phys.Lett.B:** 04 July 2024

2.6.1.2 Quantum detection of new physics in top-quark pair production at the LHC [35]

- **Authors:** Fabio Maltoni, Claudio Severi, Simone Tentori, Eleni Vryonidou
- **Posted on arXiv:** 16 January 2024
- **Published in JHEP:** 15 March 2024

2.6.1.3 Bell inequalities and quantum entanglement in weak gauge boson production at the LHC and future colliders [36]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 01 February 2023
- **Published in Eur.Phys.J.C:** 15 September 2023

2.6.1.4 Quantum Discord and Steering in Top Quarks at the LHC [37]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 08 September 2022
- **Published in Phys. Rev. Lett. 130, 221801 (2023):** 31 May 2023

2.6.1.5 Constraining new physics in entangled two-qubit systems: top-quark, tau-lepton and photon pairs [38]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli
- **Posted on arXiv:** 24 August 2022
- **Published in Eur.Phys.J.C:** 19 February 2023

2.6.1.6 Improved tests of entanglement and Bell inequalities with LHC tops [39]

- **Authors:** J.A. Aguilar-Saavedra, J.A. Casas
- **Posted on arXiv:** 01 May 2022
- **Published in Eur.Phys.J.C:** 03 August 2022

2.6.1.7 Quantum information with top quarks in QCD [40]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 10 March 2022
- **Published in Quantum:** 29 September 2022

2.6.1.8 Quantum SMEFT tomography: Top quark pair production at the LHC [41]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

2.6.1.9 Quantum tops at the LHC: from entanglement to Bell inequalities [42]

- **Authors:** Claudio Severi, Cristian Degli Esposti Boschi, Fabio Maltoni, Maximiliano Sioli
- **Posted on arXiv:** 19 October 2021
- **Published in Eur.Phys.J.C:** 02 April 2022

2.6.1.10 Testing Bell Inequalities at the LHC with Top-Quark Pairs [43]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett. 127 (2021) 16, 161801:** 15 October 2021

2.6.1.11 Entanglement and quantum tomography with top quarks at the LHC [44]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

2.7 Event Classification

2.7.1 Quantum Annealing

2.7.1.1 Quantum adiabatic machine learning by zooming into a region of the energy surface [45]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

2.7.1.2 Solving a Higgs optimization problem with quantum annealing for machine learning [46]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

2.7.2 Quantum Kernel Methods

2.7.2.1 Application of quantum machine learning in a Higgs physics study at the CEPC [47]

- **Authors:** Abdualazem Fadol, Qiyu Sha, Yaquan Fang, Zhan Li, Sitian Qian, Yuyang Xiao, Yu Zhang, Chen Zhou
- **Posted on arXiv:** 26 September 2022
- **Published in International Journal of Modern Physics A, 2024:** 07 March 2024

2.7.2.2 Higgs analysis with quantum classifiers [48]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

2.7.2.3 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [49]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research 3, 033221 (2021):** 08 September 2021

2.7.2.4 Quantum Support Vector Machines for Continuum Suppression in B Meson Decays [50]

- **Authors:** Jamie Heredge, Charles Hill, Lloyd Hollenberg, Martin Sevior
- **Posted on arXiv:** 22 March 2021
- **Published in Comput.Softw.Big Sci.:** 30 November 2021

2.7.3 Quantum Neural Networks

2.7.3.1 Enforcing exact permutation and rotational symmetries in the application of quantum neural networks on point cloud datasets [51]

- **Authors:** Zhelun Li, Lento Nagano, Koji Terashi
- **Posted on arXiv:** 17 May 2024
- **Published in Phys.Rev.Res.:** 10 October 2024

2.7.3.2 Quantum data learning for quantum simulations in high-energy physics [52]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

2.7.4 Variational Quantum Circuits

2.7.4.1 Quantum Vision Transformers for Quark–Gluon Classification [53]

- **Authors:** Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva, Eyup B. Unlu
- **Posted on arXiv:** 16 May 2024
- **Published in Axioms 2024, 13(5), 323:** 13 May 2024

2.7.4.2 Quantum Metric Learning for New Physics Searches at the LHC [54]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

2.7.4.3 Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets [27]

- **Authors:** Miguel Caçador Peixoto, Nuno Filipe Castro, Miguel Crispim Romão, Maria Gabriela Jordão Oliveira, Inês Ochoa
- **Posted on arXiv:** 06 November 2022
- **Published in Front. Artif. Intell. 6 (2023) 1268852:** 20 November 2023

2.7.4.4 Higgs analysis with quantum classifiers [48]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

2.7.4.5 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [55]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

2.7.4.6 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [56]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

2.7.4.7 Event Classification with Quantum Machine Learning in High-Energy Physics [57]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

2.8 Event Generation

2.8.1 Quantum Circuit Born Machines

2.8.1.1 Unsupervised quantum circuit learning in high energy physics [58]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton
- **Posted on arXiv:** 07 March 2022
- **Published in Phys.Rev.D:** 01 November 2022

2.8.2 Quantum Generative Adversarial Networks

2.8.2.1 Generative invertible quantum neural networks [59]

- **Authors:** Armand Rousselot, Michael Spannowsky
- **Posted on arXiv:** 24 February 2023
- **Published in SciPost Phys. 16, 146 (2024):** 04 June 2024

2.8.2.2 Quantum integration of elementary particle processes [60]

- **Authors:** Gabriele Agliardi, Michele Grossi, Mathieu Pellen, Enrico Prati
- **Posted on arXiv:** 05 January 2022
- **Published in Phys.Lett.B:** 07 June 2022

2.8.2.3 Style-based quantum generative adversarial networks for Monte Carlo events [61]

- **Authors:** Carlos Bravo-Prieto, Julien Baglio, Marco Cè, Anthony Francis, Dorota M. Grabowska, Stefano Carrazza
- **Posted on arXiv:** 13 October 2021
- **Published in Quantum 6, 777 (2022):** 17 August 2022

2.8.3 Quantum Information Theory

2.8.3.1 Three-Body Entanglement in Particle Decays [62]

- **Authors:** Kazuki Sakurai, Michael Spannowsky
- **Posted on arXiv:** 02 October 2023
- **Published in Phys.Rev.Lett.:** 09 April 2024

2.8.4 Quantum Simulations

2.8.4.1 Towards a quantum computing algorithm for helicity amplitudes and parton showers [63]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 13 October 2020
- **Published in Phys. Rev. D 103, 076020 (2021):** 27 April 2021

2.8.4.2 Quantum Algorithm for High Energy Physics Simulations [64]

- **Authors:** Christian W. Bauer, Wibe A. de Jong, Benjamin Nachman, Davide Provasoli
- **Posted on arXiv:** 05 April 2019
- **Published in Phys. Rev. Lett. 126, 062001 (2021):** 11 February 2021

2.8.5 Quantum Walks

2.8.5.1 Collider events on a quantum computer [65]

- **Authors:** Gösta Gustafson, Stefan Prestel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 21 July 2022
- **Published in JHEP:** 07 November 2022

2.8.5.2 Quantum walk approach to simulating parton showers [66]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 28 September 2021
- **Published in Phys. Rev. D 106 (2022) 056002:** 01 September 2022

2.8.6 Variational Quantum Circuits

2.8.6.1 Partonic collinear structure by quantum computing [67]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.9 Jet Algorithms and Jet Tagging

2.9.1 Quantum Algorithms Based on Amplitude Amplification

2.9.1.1 Quantum Algorithms in Particle Physics [68]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

2.9.1.2 Quantum Algorithms for Jet Clustering [69]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

2.9.2 Quantum Annealing

2.9.2.1 Degeneracy engineering for classical and quantum annealing: A case study of sparse linear regression in collider physics [70]

- **Authors:** Eric R. Anschuetz, Lena Funcke, Patrick T. Komiske, Serhii Kryhin, Jesse Thaler
- **Posted on arXiv:** 20 May 2022
- **Published in Phys. Rev. D 106, 056008 (2022):** 01 September 2022

2.9.2.2 Quantum annealing for jet clustering with thrust [71]

- **Authors:** Andrea Delgado, Jesse Thaler
- **Posted on arXiv:** 05 May 2022
- **Published in Phys.Rev.D:** 01 November 2022

2.9.2.3 Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders [72]

- **Authors:** Minho Kim, Pyungwon Ko, Jae-hyeon Park, Myeonghun Park
- **Posted on arXiv:** 15 November 2021

2.9.2.4 Adiabatic quantum algorithm for multijet clustering in high energy physics [73]

- **Authors:** Diogo Pires, Yasser Omar, João Seixas
- **Posted on arXiv:** 28 December 2020
- **Published in Phys.Lett.B:** 13 June 2023

2.9.2.5 Quantum Algorithms for Jet Clustering [69]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

2.9.3 Quantum Inspired Algorithms

2.9.3.1 Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States [74]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 15 June 2021
- **Published in JHEP 08 (2021) 112:** 23 August 2021

2.9.3.2 Quantum-inspired machine learning on high-energy physics data [75]

- **Authors:** Timo Felser, Marco Trenti, Lorenzo Sestini, Alessio Gianelle, Davide Zuliani, Donatella Lucchesi, Simone Montangero
- **Posted on arXiv:** 28 April 2020
- **Published in npj Quantum Inf.:** 15 July 2021

2.9.4 Quantum Unsupervised Clustering Algorithms

2.9.4.1 Quantum clustering and jet reconstruction at the LHC [76]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

2.9.4.2 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [77]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

2.9.5 Tensor Networks

2.9.5.1 Classical versus quantum: Comparing tensor-network-based quantum circuits on Large Hadron Collider data [78]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 21 February 2022
- **Published in Phys. Rev. A 106 (Dec, 2022) 062423:** 19 December 2022

2.9.5.2 Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States [74]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 15 June 2021
- **Published in JHEP 08 (2021) 112:** 23 August 2021

2.9.6 Variational Quantum Circuits

2.9.6.1 Quantum Machine Learning for b-jet charge identification [79]

- **Authors:** Alessio Gianelle, Patrick Koppenburg, Donatella Lucchesi, Davide Nicotra, Eduardo Rodrigues, Lorenzo Sestini, Jacco de Vries, Davide Zuliani
- **Posted on arXiv:** 28 February 2022
- **Published in JHEP:** 01 August 2022

2.10 Lattice Field Theories

2.10.1 Quantum Annealing

2.10.1.1 SU(2) lattice gauge theory on a quantum annealer [80]

- **Authors:** Sarmed A Rahman, Randy Lewis, Emanuele Mendicelli, Sarah Powell
- **Posted on arXiv:** 15 March 2021
- **Published in Phys. Rev. D 104, 034501 (2021):** 01 August 2021

2.10.1.2 A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer [81]

- **Authors:** Nga T.T. Nguyen, Garrett T. Kenyon, Boram Yoon
- **Posted on arXiv:** 14 November 2019
- **Published in Sci Rep 10, 10915 (2020):** 02 July 2020

2.10.2 Quantum Simulations

2.10.2.1 Steps toward quantum simulations of hadronization and energy loss in dense matter [82]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

2.10.2.2 Efficient representation for simulating U(1) gauge theories on digital quantum computers at all values of the coupling [83]

- **Authors:** Christian W. Bauer, Dorota M. Grabowska
- **Posted on arXiv:** 15 November 2021
- **Published in Phys.Rev.D:** 01 February 2023

2.10.2.3 Lattice renormalization of quantum simulations [84]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 02 July 2021
- **Published in Phys.Rev.D:** 01 November 2021

2.10.2.4 SU(2) hadrons on a quantum computer via a variational approach [85]

- **Authors:** Yasar Y. Atas, Jinglei Zhang, Randy Lewis, Amin Jahanpour, Jan F. Haase, Christine A. Muschik
- **Posted on arXiv:** 17 February 2021
- **Published in Nature Communications 2021:** 11 November 2021

2.10.2.5 Quantum simulation of gauge theory via orbifold lattice [86]

- **Authors:** Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu
- **Posted on arXiv:** 12 November 2020
- **Published in JHEP 2109 (2021) 034:** 06 September 2021

2.10.2.6 Role of boundary conditions in quantum computations of scattering observables [87]

- **Authors:** Raúl A. Briceño, Juan V. Guerrero, Maxwell T. Hansen, Alexandru M. Sturzu
- **Posted on arXiv:** 01 July 2020
- **Published in Phys. Rev. D 103, 014506 (2021):** 07 January 2021

2.10.2.7 Toward scalable simulations of Lattice Gauge Theories on quantum computers [88]

- **Authors:** Simon V. Mathis, Guglielmo Mazzola, Ivano Tavernelli
- **Posted on arXiv:** 20 May 2020
- **Published in Phys. Rev. D 102, 094501 (2020):** 04 November 2020

2.10.2.8 Simulating Lattice Gauge Theories within Quantum Technologies [89]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

2.10.2.9 Simulating lattice gauge theories on a quantum computer [90]

- **Authors:** Tim Byrnes, Yoshihisa Yamamoto
- **Posted on arXiv:** October 2005
- **Published in Phys.Rev.A:** 2006

2.11 Neutrinos

2.11.1 Quantum Simulations

2.11.1.1 Neutrino Oscillations in a Quantum Processor [91]

- **Authors:** C.A. Argüelles, B.J. P. Jones
- **Posted on arXiv:** 23 April 2019
- **Published in Phys. Rev. Research 1, 033176 (2019):** 13 December 2019

2.11.2 Variational Quantum Circuits

2.11.2.1 Hybrid Quantum-Classical Graph Convolutional Network [92]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

2.11.2.2 Quantum convolutional neural networks for high energy physics data analysis [93]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

2.12 Quantum Field Theories

2.12.1 Quantum Algorithms Based on Amplitude Amplification

2.12.1.1 Quantum Algorithms in Particle Physics [68]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

2.12.1.2 Quantum algorithm for Feynman loop integrals [94]

- **Authors:** Selomit Ramírez-Uribe, Andrès E. Rentería-Olivo, Germán Rodrigo, German F.R. Sborlini, Luiz Vale Silva
- **Posted on arXiv:** 18 May 2021
- **Published in JHEP:** 16 May 2022

2.12.2 Quantum Annealing

2.12.2.1 Preparations for quantum simulations of quantum chromodynamics in <math display=”inline”><mrow><mn>1</mn><mo>+</mo><mn>1</mn></mrow></ dimensions. I. Axial gauge [95]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

2.12.3 Quantum Information Theory

2.12.3.1 Can Bell inequalities be tested via scattering cross-section at colliders ? [96]

- **Authors:** Song Li, Wei Shen, Jin Min Yang
- **Posted on arXiv:** 02 January 2024
- **Published in Eur.Phys.J.C:** 21 November 2024

2.12.3.2 Minimal entanglement and emergent symmetries in low-energy QCD [97]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

2.12.3.3 Symmetry from entanglement suppression [98]

- **Authors:** Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 April 2021
- **Published in Phys.Rev.D:** 01 October 2021

2.12.4 Quantum Simulations

2.12.4.1 Quantum Simulation of SU(3) Lattice Yang-Mills Theory at Leading Order in Large-<math display=”inline”><msub><mi>N</mi><mi>c</mi></msub></math> Expansion [99]

- **Authors:** Anthony N. Ciavarella, Christian W. Bauer
- **Posted on arXiv:** 15 February 2024
- **Published in Phys.Rev.Lett.:** 09 September 2024

2.12.4.2 Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics [100]

- **Authors:** Eyup B. Unlu, Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhong-tian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva
- **Posted on arXiv:** 01 February 2024
- **Published in Axioms v. 13, no 3, (2024) 187:** 13 March 2024

2.12.4.3 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [101]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

2.12.4.4 Toward QCD on quantum computer: orbifold lattice approach [102]

- **Authors:** Georg Bergner, Masanori Hanada, Enrico Rinaldi, Andreas Schafer
- **Posted on arXiv:** 22 January 2024
- **Published in JHEP:** 20 May 2024

2.12.4.5 Loop Feynman integration on a quantum computer [103]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Michele Grossi, Sofia Vallecorsa, Germán Rodrigo
- **Posted on arXiv:** 05 January 2024
- **Published in Phys. Rev. D 110, 074031 (2024):** 01 October 2024

2.12.4.6 Quantum simulation of fundamental particles and forces [104]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 09 April 2024
- **Published in Nature Rev.Phys.:** 21 June 2023

2.12.4.7 Quantum simulation of colour in perturbative quantum chromodynamics [105]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen
- **Posted on arXiv:** 08 March 2023
- **Published in SciPost Phys. 15, 205 (2023):** 24 November 2023

2.12.4.8 Preparations for quantum simulations of quantum chromodynamics in <math display=”inline”><mrow><mn>1</mn><mo>+</mo><mn>1</mn></mrow></ dimensions. I. Axial gauge [95]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

2.12.4.9 Simulating Collider Physics on Quantum Computers Using Effective Field Theories [106]

- **Authors:** Christian W. Bauer, Marat Freytsis, Benjamin Nachman
- **Posted on arXiv:** 09 February 2021
- **Published in Phys.Rev.Lett.:** 18 November 2021

2.12.4.10 Quantum simulation of quantum field theory in the light-front formulation [107]

- **Authors:** Michael Kreshchuk, William M. Kirby, Gary Goldstein, Hugo Beauchemin, Peter J. Love
- **Posted on arXiv:** 10 February 2020
- **Published in Phys. Rev. A 105, 032418 (2022):** 08 March 2022

2.12.4.11 SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers [108]

- **Authors:** Natalie Klco, Jesse R. Stryker, Martin J. Savage
- **Posted on arXiv:** 19 August 2019
- **Published in Phys. Rev. D 101, 074512 (2020):** 22 April 2020

2.12.4.12 General Methods for Digital Quantum Simulation of Gauge Theories [109]

- **Authors:** Henry Lamm, Scott Lawrence, Yukari Yamauchi
- **Posted on arXiv:** 19 March 2019
- **Published in Phys. Rev. D 100, 034518 (2019):** 29 August 2019

2.12.4.13 Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers [110]

- **Authors:** Kubra Yeter-Aydeniz, Eugene F. Dumitrescu, Alex J. McCaskey, Ryan S. Bennink, Raphael C. Pooser, George Siopsis
- **Posted on arXiv:** 29 November 2018
- **Published in Phys. Rev. A 99, 032306 (2019):** 04 March 2019

2.12.4.14 Digitization of scalar fields for quantum computing [111]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 30 August 2018
- **Published in Phys. Rev. A 99, 052335 (2019):** 23 May 2019

2.12.4.15 Real-time dynamics of lattice gauge theories with a few-qubit quantum computer [112]

- **Authors:** E.A. Martinez, C.A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmonte, T. Monz, P. Zoller, R. Blatt
- **Posted on arXiv:** 15 May 2016
- **Published in Nature:** 22 June 2016

2.12.4.16 Towards Quantum Simulating QCD [113]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 25 September 2014
- **Published in Nucl.Phys.A:** 28 September 2014

2.12.4.17 Quantum Algorithms for Fermionic Quantum Field Theories [114]

- **Authors:** Stephen P. Jordan, Keith S. M. Lee, John Preskill
- **Posted on arXiv:** 28 April 2014

2.12.4.18 Quantum Computation of Scattering in Scalar Quantum Field Theories [115]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** December 2011

2.12.5 Variational Quantum Circuits

2.12.5.1 Quantum-classical simulation of quantum field theory by quantum circuit learning [116]

- **Authors:** Kazuki Ikeda
- **Posted on arXiv:** 27 November 2023

2.13 Track Reconstruction

2.13.1 Quantum Algorithms Based on Amplitude Amplification

2.13.1.1 Quantum speedup for track reconstruction in particle accelerators [117]

- **Authors:** Duarte Magano, Akshat Kumar, Mārtiņš Kālis, Andris Locāns, Adam Glos, Sagar Pratapsi, Gonçalo Quinta, Maksims Dimitrijevs, Aleksander Rivošs, Pedrame Bargassa, João Seixas, Andris Ambainis, Yasser Omar
- **Posted on arXiv:** 23 April 2021
- **Published in Physical Review D 105 (2022) 076012:** 01 April 2022

2.13.2 Quantum Annealing

2.13.2.1 Particle track classification using quantum associative memory [118]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

2.13.2.2 Quantum annealing algorithms for track pattern recognition [119]

- **Authors:** Masahiko Saito, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder, Yasuyuki Okumura, Ryu Sawada, Alex Smith, Junichi Tanaka, Koji Terashi
- **Published in EPJ Web Conf.:** 2020

2.13.2.3 Charged particle tracking with quantum annealing optimization [120]

- **Authors:** Alexander Zlokapa, Abhishek Anand, Jean-Roch Vlimant, Javier M. Duarte, Joshua Job, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 12 August 2019
- **Published in Quantum Mach. Intell. 3, 27 (2021):** 02 November 2021

2.13.2.4 Track clustering with a quantum annealer for primary vertex reconstruction at hadron colliders [121]

- **Authors:** Souvik Das, Andrew J. Wildridge, Sachin B. Vaidya, Andreas Jung
- **Posted on arXiv:** 21 March 2019

2.13.2.5 A pattern recognition algorithm for quantum annealers [122]

- **Authors:** Frèdèric Bapst, Wahid Bhimji, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder
- **Posted on arXiv:** 21 February 2019
- **Published in Comput.Softw.Big Sci.:** 09 December 2019

2.13.3 Quantum Kernel Methods

2.13.3.1 Reconstructing charged particle track segments with a quantum-enhanced support vector machine [123]

- **Authors:** Philippa Duckett, Gabriel Facini, Marcin Jastrzebski, Sarah Malik, Tim Scanlon, Sebastien Rettie
- **Posted on arXiv:** 14 December 2022
- **Published in Phys.Rev.D:** 01 March 2024

2.13.4 Quantum Neural Networks

2.13.4.1 Hybrid quantum classical graph neural networks for particle track reconstruction [124]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

2.13.5 Quantum Storage

2.13.5.1 Particle track classification using quantum associative memory [118]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

2.13.5.2 Quantum Associative Memory in HEP Track Pattern Recognition [125]

- **Authors:** Illya Shapoval, Paolo Calafiura
- **Posted on arXiv:** 29 January 2019
- **Published in EPJ Web of Conferences 214 (2019) 01012:** 2019

2.13.6 Variational Quantum Circuits

2.13.6.1 Quantum-Annealing-Inspired Algorithms for Track Reconstruction at High-Energy Colliders [126]

- **Authors:** Hideki Okawa, Qing-Guo Zeng, Xian-Zhe Tao, Man-Hong Yung
- **Posted on arXiv:** 22 February 2024
- **Published in Comput.Softw.Big Sci.:** 28 August 2024

2.13.6.2 Charged particle reconstruction for future high energy colliders with Quantum Approximate Optimization Algorithm [127]

- **Authors:** Hideki Okawa
- **Posted on arXiv:** 16 October 2023

2.13.6.3 Particle track reconstruction with noisy intermediate-scale quantum computers [128]

- **Authors:** Tim Schwägerl, Cigdem Issever, Karl Jansen, Teng Jian Khoo, Stefan Kühn, Cenk Tüysüz, Hannsjörg Weber
- **Posted on arXiv:** 23 March 2023

2.13.6.4 Studying quantum algorithms for particle track reconstruction in the LUXE experiment [129]

- **Authors:** Lena Funcke, Tobias Hartung, Beate Heinemann, Karl Jansen, Annabel Kropf, Stefan Kühn, Federico Meloni, David Spataro, Cenk Tüysüz, Yee Chinn Yap
- **Posted on arXiv:** 14 February 2022
- **Published in J.Phys.Conf.Ser.:** 2023

3 Quantum Information Science in Nuclear and Particle Physics

3.1 Reviews and Whitepapers

3.1.1 Reviews

3.1.1.1 Quantum entanglement and Bell inequality violation at colliders [1]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

3.1.1.2 Quantum Computing Applications in Future Colliders [2]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

3.1.1.3 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 2021

3.1.2 Whitepapers

3.1.2.1 Quantum Computing for High-Energy Physics: State of the Art and Challenges [4]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.1.2.2 Report of the Snowmass 2021 Theory Frontier Topical Group on Quantum Information Science [5]

- **Authors:** Simon Catterall, Roni Harnik, Veronika E. Hubeny, Christian W. Bauer, Asher Berlin, Zohreh Davoudi, Thomas Faulkner, Thomas Hartman, Matthew Headrick, Yonatan F. Kahn, Henry Lamm, Yannick Meurice, Surjeet Rajendran, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 29 September 2022

3.1.2.3 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [6]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.1.2.4 Quantum computing hardware for HEP algorithms and sensing [7]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Cancelo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti
- **Posted on arXiv:** 18 April 2022

3.1.2.5 Quantum Simulation for High-Energy Physics [8]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

3.1.2.6 Quantum Networks for High Energy Physics [9]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie
- **Posted on arXiv:** 31 March 2022

3.1.2.7 New Horizons: Scalar and Vector Ultralight Dark Matter [10]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou
- **Posted on arXiv:** 28 March 2022

3.1.2.8 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.1.2.9 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [12]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

3.1.2.10 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [13]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

3.1.2.11 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [14]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

3.2 Continuous Variable Quantum Computing

3.2.1 Anomaly Detection

3.2.1.1 Unsupervised event classification with graphs on classical and photonic quantum computers [16]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

3.2.2 Detector Simulation

3.2.2.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [29]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

3.3 Quantum Algorithms Based on Amplitude Amplification

3.3.1 Beyond the Standard Model

3.3.1.1 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [22]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

3.3.1.2 Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider [23]

- **Authors:** Anthony E. Armenakas, Oliver K. Baker
- **Posted on arXiv:** 01 October 2020

3.3.2 Jet Algorithms and Jet Tagging

3.3.2.1 Quantum Algorithms in Particle Physics [68]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

3.3.2.2 Quantum Algorithms for Jet Clustering [69]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

3.3.3 Quantum Field Theories

3.3.3.1 Quantum Algorithms in Particle Physics [68]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

3.3.3.2 Quantum algorithm for Feynman loop integrals [94]

- **Authors:** Selomit Ramírez-Uribe, Andrès E. Rentería-Olivo, Germán Rodrigo, German F.R. Sborlini, Luiz Vale Silva
- **Posted on arXiv:** 18 May 2021
- **Published in JHEP:** 16 May 2022

3.3.4 Track Reconstruction

3.3.4.1 Quantum speedup for track reconstruction in particle accelerators [117]

- **Authors:** Duarte Magano, Akshat Kumar, Mārtiņš Kālis, Andris Locāns, Adam Glos, Sagar Pratapsi, Gonçalo Quinta, Maksims Dimitrijevs, Aleksander Rivošs, Pedrame Bargassa, João Seixas, Andris Ambainis, Yasser Omar
- **Posted on arXiv:** 23 April 2021
- **Published in Physical Review D 105 (2022) 076012:** 01 April 2022

3.4 Quantum Annealing

3.4.1 Anomaly Detection

3.4.1.1 A Quantum Algorithm for Model-Independent Searches for New Physics [17]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

3.4.2 Beyond the Standard Model

3.4.2.1 Completely quantum neural networks [24]

- **Authors:** Steve Abel, Juan C. Criado, Michael Spannowsky
- **Posted on arXiv:** 23 February 2022
- **Published in Phys.Rev.A:** 01 August 2022

3.4.2.2 Quantum algorithm for the classification of supersymmetric top quark events [25]

- **Authors:** Pedrame Bargassa, Timothèe Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothèe Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

3.4.3 Cosmology and Early Universe Physics

3.4.3.1 Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer [28]

- **Authors:** João Caldeira, Joshua Job, Steven H. Adachi, Brian Nord, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019

3.4.4 Detector Simulation

3.4.4.1 Hybrid actor-critic algorithm for quantum reinforcement learning at CERN beam lines [30]

- **Authors:** Michael Schenk, Elías F. Combarro, Michele Grossi, Verena Kain, Kevin Shing Bruce Li, Mircea-Marian Popa, Sofia Vallecorsa
- **Posted on arXiv:** 22 September 2022
- **Published in Quantum Sci.Technol.:** 21 February 2024

3.4.5 Event Classification

3.4.5.1 Quantum adiabatic machine learning by zooming into a region of the energy surface [45]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

3.4.5.2 Solving a Higgs optimization problem with quantum annealing for machine learning [46]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

3.4.6 Jet Algorithms and Jet Tagging

3.4.6.1 Degeneracy engineering for classical and quantum annealing: A case study of sparse linear regression in collider physics [70]

- **Authors:** Eric R. Anschuetz, Lena Funcke, Patrick T. Komiske, Serhii Kryhin, Jesse Thaler
- **Posted on arXiv:** 20 May 2022
- **Published in Phys. Rev. D 106, 056008 (2022):** 01 September 2022

3.4.6.2 Quantum annealing for jet clustering with thrust [71]

- **Authors:** Andrea Delgado, Jesse Thaler
- **Posted on arXiv:** 05 May 2022
- **Published in Phys.Rev.D:** 01 November 2022

3.4.6.3 Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders [72]

- **Authors:** Minho Kim, Pyungwon Ko, Jae-hyeon Park, Myeonghun Park
- **Posted on arXiv:** 15 November 2021

3.4.6.4 Adiabatic quantum algorithm for multijet clustering in high energy physics [73]

- **Authors:** Diogo Pires, Yasser Omar, João Seixas
- **Posted on arXiv:** 28 December 2020
- **Published in Phys.Lett.B:** 13 June 2023

3.4.6.5 Quantum Algorithms for Jet Clustering [69]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

3.4.7 Lattice Field Theories

3.4.7.1 SU(2) lattice gauge theory on a quantum annealer [80]

- **Authors:** Sarmed A Rahman, Randy Lewis, Emanuele Mendicelli, Sarah Powell
- **Posted on arXiv:** 15 March 2021
- **Published in Phys. Rev. D 104, 034501 (2021):** 01 August 2021

3.4.7.2 A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer [81]

- **Authors:** Nga T.T. Nguyen, Garrett T. Kenyon, Boram Yoon
- **Posted on arXiv:** 14 November 2019
- **Published in Sci Rep 10, 10915 (2020):** 02 July 2020

3.4.8 Quantum Field Theories

3.4.8.1 Preparations for quantum simulations of quantum chromodynamics in <math display=”inline”><mrow><mn>1</mn><mo>+</mo><mn>1</mn></mrow></ dimensions. I. Axial gauge [95]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

3.4.9 Track Reconstruction

3.4.9.1 Particle track classification using quantum associative memory [118]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

3.4.9.2 Quantum annealing algorithms for track pattern recognition [119]

- **Authors:** Masahiko Saito, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder, Yasuyuki Okumura, Ryu Sawada, Alex Smith, Junichi Tanaka, Koji Terashi
- **Published in EPJ Web Conf.:** 2020

3.4.9.3 Charged particle tracking with quantum annealing optimization [120]

- **Authors:** Alexander Zlokapa, Abhishek Anand, Jean-Roch Vlimant, Javier M. Duarte, Joshua Job, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 12 August 2019
- **Published in Quantum Mach. Intell. 3, 27 (2021):** 02 November 2021

3.4.9.4 Track clustering with a quantum annealer for primary vertex reconstruction at hadron colliders [121]

- **Authors:** Souvik Das, Andrew J. Wildridge, Sachin B. Vaidya, Andreas Jung
- **Posted on arXiv:** 21 March 2019

3.4.9.5 A pattern recognition algorithm for quantum annealers [122]

- **Authors:** Frèdèric Bapst, Wahid Bhimji, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder
- **Posted on arXiv:** 21 February 2019
- **Published in Comput.Softw.Big Sci.:** 09 December 2019

3.5 Quantum Autoencoders

3.5.1 Anomaly Detection

3.5.1.1 Anomaly detection in high-energy physics using a quantum autoencoder [18]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi
- **Posted on arXiv:** 09 December 2021
- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

3.6 Quantum Circuit Born Machines

3.6.1 Event Generation

3.6.1.1 Unsupervised quantum circuit learning in high energy physics [58]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton
- **Posted on arXiv:** 07 March 2022
- **Published in Phys.Rev.D:** 01 November 2022

3.7 Quantum Kernel Methods

3.7.1 Anomaly Detection

3.7.1.1 Quantum anomaly detection in the latent space of proton collision events at the LHC [19]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

3.7.1.2 Unravelling physics beyond the standard model with classical and quantum anomaly detection [20]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

3.7.2 Beyond the Standard Model

3.7.2.1 Unravelling physics beyond the standard model with classical and quantum anomaly detection [20]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

3.7.3 Event Classification

3.7.3.1 Application of quantum machine learning in a Higgs physics study at the CEPC [47]

- **Authors:** Abdualazem Fadol, Qiyu Sha, Yaquan Fang, Zhan Li, Sitian Qian, Yuyang Xiao, Yu Zhang, Chen Zhou
- **Posted on arXiv:** 26 September 2022
- **Published in International Journal of Modern Physics A, 2024:** 07 March 2024

3.7.3.2 Higgs analysis with quantum classifiers [48]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

3.7.3.3 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [49]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research 3, 033221 (2021):** 08 September 2021

3.7.3.4 Quantum Support Vector Machines for Continuum Suppression in B Meson Decays [50]

- **Authors:** Jamie Heredge, Charles Hill, Lloyd Hollenberg, Martin Sevior
- **Posted on arXiv:** 22 March 2021
- **Published in Comput.Softw.Big Sci.:** 30 November 2021

3.7.4 Track Reconstruction

3.7.4.1 Reconstructing charged particle track segments with a quantum-enhanced support vector machine [123]

- **Authors:** Philippa Duckett, Gabriel Facini, Marcin Jastrzebski, Sarah Malik, Tim Scanlon, Sebastien Rettie
- **Posted on arXiv:** 14 December 2022
- **Published in Phys.Rev.D:** 01 March 2024

3.8 Quantum Neural Networks

3.8.1 Event Classification

3.8.1.1 Enforcing exact permutation and rotational symmetries in the application of quantum neural networks on point cloud datasets [51]

- **Authors:** Zhelun Li, Lento Nagano, Koji Terashi
- **Posted on arXiv:** 17 May 2024
- **Published in Phys.Rev.Res.:** 10 October 2024

3.8.1.2 Quantum data learning for quantum simulations in high-energy physics [52]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

3.8.2 Track Reconstruction

3.8.2.1 Hybrid quantum classical graph neural networks for particle track reconstruction [124]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

3.9 Quantum Generative Adversarial Networks

3.9.1 Detector Simulation

3.9.1.1 Running the Dual-PQC GAN on noisy simulators and real quantum hardware [31]

- **Authors:** Su Yeon Chang, Edwin Agnew, Elías F. Combarro, Michele Grossi, Steven Herbert, Sofia Vallecorsa
- **Posted on arXiv:** 30 May 2022
- **Published in J.Phys.Conf.Ser.:** 2023

3.9.1.2 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [32]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

3.9.1.3 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [29]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

3.9.2 Event Generation

3.9.2.1 Generative invertible quantum neural networks [59]

- **Authors:** Armand Rousselot, Michael Spannowsky
- **Posted on arXiv:** 24 February 2023
- **Published in SciPost Phys. 16, 146 (2024):** 04 June 2024

3.9.2.2 Quantum integration of elementary particle processes [60]

- **Authors:** Gabriele Agliardi, Michele Grossi, Mathieu Pellen, Enrico Prati
- **Posted on arXiv:** 05 January 2022
- **Published in Phys.Lett.B:** 07 June 2022

3.9.2.3 Style-based quantum generative adversarial networks for Monte Carlo events [61]

- **Authors:** Carlos Bravo-Prieto, Julien Baglio, Marco Cè, Anthony Francis, Dorota M. Grabowska, Stefano Carrazza
- **Posted on arXiv:** 13 October 2021
- **Published in Quantum 6, 777 (2022):** 17 August 2022

3.10 Quantum Inspired Algorithms

3.10.1 Jet Algorithms and Jet Tagging

3.10.1.1 Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States [74]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 15 June 2021
- **Published in JHEP 08 (2021) 112:** 23 August 2021

3.10.1.2 Quantum-inspired machine learning on high-energy physics data [75]

- **Authors:** Timo Felser, Marco Trenti, Lorenzo Sestini, Alessio Gianelle, Davide Zuliani, Donatella Lucchesi, Simone Montangero
- **Posted on arXiv:** 28 April 2020
- **Published in npj Quantum Inf.:** 15 July 2021

3.11 Quantum Information Theory

3.11.1 Elementary Particle Systems

3.11.1.1 Full quantum tomography of top quark decays [34]

- **Authors:** J.A. Aguilar-Saavedra
- **Posted on arXiv:** 22 February 2024
- **Published in Phys.Lett.B:** 04 July 2024

3.11.1.2 Quantum detection of new physics in top-quark pair production at the LHC [35]

- **Authors:** Fabio Maltoni, Claudio Severi, Simone Tentori, Eleni Vryonidou
- **Posted on arXiv:** 16 January 2024
- **Published in JHEP:** 15 March 2024

3.11.1.3 Bell inequalities and quantum entanglement in weak gauge boson production at the LHC and future colliders [36]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 01 February 2023
- **Published in Eur.Phys.J.C:** 15 September 2023

3.11.1.4 Quantum Discord and Steering in Top Quarks at the LHC [37]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 08 September 2022
- **Published in Phys. Rev. Lett. 130, 221801 (2023):** 31 May 2023

3.11.1.5 Constraining new physics in entangled two-qubit systems: top-quark, tau-lepton and photon pairs [38]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli
- **Posted on arXiv:** 24 August 2022
- **Published in Eur.Phys.J.C:** 19 February 2023

3.11.1.6 Improved tests of entanglement and Bell inequalities with LHC tops [39]

- **Authors:** J.A. Aguilar-Saavedra, J.A. Casas
- **Posted on arXiv:** 01 May 2022
- **Published in Eur.Phys.J.C:** 03 August 2022

3.11.1.7 Quantum information with top quarks in QCD [40]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 10 March 2022
- **Published in Quantum:** 29 September 2022

3.11.1.8 Quantum SMEFT tomography: Top quark pair production at the LHC [41]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

3.11.1.9 Quantum tops at the LHC: from entanglement to Bell inequalities [42]

- **Authors:** Claudio Severi, Cristian Degli Esposti Boschi, Fabio Maltoni, Maximiliano Sioli
- **Posted on arXiv:** 19 October 2021
- **Published in Eur.Phys.J.C:** 02 April 2022

3.11.1.10 Testing Bell Inequalities at the LHC with Top-Quark Pairs [43]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett. 127 (2021) 16, 161801:** 15 October 2021

3.11.1.11 Entanglement and quantum tomography with top quarks at the LHC [44]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

3.11.2 Event Generation

3.11.2.1 Three-Body Entanglement in Particle Decays [62]

- **Authors:** Kazuki Sakurai, Michael Spannowsky
- **Posted on arXiv:** 02 October 2023
- **Published in Phys.Rev.Lett.:** 09 April 2024

3.11.3 Quantum Field Theories

3.11.3.1 Can Bell inequalities be tested via scattering cross-section at colliders ? [96]

- **Authors:** Song Li, Wei Shen, Jin Min Yang
- **Posted on arXiv:** 02 January 2024
- **Published in Eur.Phys.J.C:** 21 November 2024

3.11.3.2 Minimal entanglement and emergent symmetries in low-energy QCD [97]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

3.11.3.3 Symmetry from entanglement suppression [98]

- **Authors:** Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 April 2021
- **Published in Phys.Rev.D:** 01 October 2021

3.12 Quantum Sensors

3.12.1 Beyond the Standard Model

3.12.1.1 Searching for Dark Matter with a Superconducting Qubit [26]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

3.13 Quantum Simulations

3.13.1 Whitepapers

3.13.1.1 Review on Quantum Computing for Lattice Field Theory [15]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in PoS:** 01 February 2023

3.13.2 Event Generation

3.13.2.1 Towards a quantum computing algorithm for helicity amplitudes and parton showers [63]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 13 October 2020
- **Published in Phys. Rev. D 103, 076020 (2021):** 27 April 2021

3.13.2.2 Quantum Algorithm for High Energy Physics Simulations [64]

- **Authors:** Christian W. Bauer, Wibe A. de Jong, Benjamin Nachman, Davide Provasoli
- **Posted on arXiv:** 05 April 2019
- **Published in Phys. Rev. Lett. 126, 062001 (2021):** 11 February 2021

3.13.3 Lattice Field Theories

3.13.3.1 Steps toward quantum simulations of hadronization and energy loss in dense matter [82]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

3.13.3.2 Efficient representation for simulating U(1) gauge theories on digital quantum computers at all values of the coupling [83]

- **Authors:** Christian W. Bauer, Dorota M. Grabowska
- **Posted on arXiv:** 15 November 2021
- **Published in Phys.Rev.D:** 01 February 2023

3.13.3.3 Lattice renormalization of quantum simulations [84]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 02 July 2021
- **Published in Phys.Rev.D:** 01 November 2021

3.13.3.4 SU(2) hadrons on a quantum computer via a variational approach [85]

- **Authors:** Yasar Y. Atas, Jinglei Zhang, Randy Lewis, Amin Jahanpour, Jan F. Haase, Christine A. Muschik
- **Posted on arXiv:** 17 February 2021
- **Published in Nature Communications 2021:** 11 November 2021

3.13.3.5 Quantum simulation of gauge theory via orbifold lattice [86]

- **Authors:** Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu
- **Posted on arXiv:** 12 November 2020
- **Published in JHEP 2109 (2021) 034:** 06 September 2021

3.13.3.6 Role of boundary conditions in quantum computations of scattering observables [87]

- **Authors:** Raúl A. Briceño, Juan V. Guerrero, Maxwell T. Hansen, Alexandru M. Sturzu
- **Posted on arXiv:** 01 July 2020
- **Published in Phys. Rev. D 103, 014506 (2021):** 07 January 2021

3.13.3.7 Toward scalable simulations of Lattice Gauge Theories on quantum computers [88]

- **Authors:** Simon V. Mathis, Guglielmo Mazzola, Ivano Tavernelli
- **Posted on arXiv:** 20 May 2020
- **Published in Phys. Rev. D 102, 094501 (2020):** 04 November 2020

3.13.3.8 Simulating Lattice Gauge Theories within Quantum Technologies [89]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.13.3.9 Simulating lattice gauge theories on a quantum computer [90]

- **Authors:** Tim Byrnes, Yoshihisa Yamamoto
- **Posted on arXiv:** October 2005
- **Published in Phys.Rev.A:** 2006

3.13.4 Neutrinos

3.13.4.1 Neutrino Oscillations in a Quantum Processor [91]

- **Authors:** C.A. Argüelles, B.J. P. Jones
- **Posted on arXiv:** 23 April 2019
- **Published in Phys. Rev. Research 1, 033176 (2019):** 13 December 2019

3.13.5 Quantum Field Theories

3.13.5.1 Quantum Simulation of SU(3) Lattice Yang-Mills Theory at Leading Order in Large-<math display=”inline”><msub><mi>N</mi><mi>c</mi></msub></math> Expansion [99]

- **Authors:** Anthony N. Ciavarella, Christian W. Bauer
- **Posted on arXiv:** 15 February 2024
- **Published in Phys.Rev.Lett.:** 09 September 2024

3.13.5.2 Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics [100]

- **Authors:** Eyup B. Unlu, Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhong-tian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva
- **Posted on arXiv:** 01 February 2024
- **Published in Axioms v. 13, no 3, (2024) 187:** 13 March 2024

3.13.5.3 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [101]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

3.13.5.4 Toward QCD on quantum computer: orbifold lattice approach [102]

- **Authors:** Georg Bergner, Masanori Hanada, Enrico Rinaldi, Andreas Schafer
- **Posted on arXiv:** 22 January 2024
- **Published in JHEP:** 20 May 2024

3.13.5.5 Loop Feynman integration on a quantum computer [103]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Michele Grossi, Sofia Vallecorsa, Germán Rodrigo
- **Posted on arXiv:** 05 January 2024
- **Published in Phys. Rev. D 110, 074031 (2024):** 01 October 2024

3.13.5.6 Quantum simulation of fundamental particles and forces [104]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 09 April 2024
- **Published in Nature Rev.Phys.:** 21 June 2023

3.13.5.7 Quantum simulation of colour in perturbative quantum chromodynamics [105]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen
- **Posted on arXiv:** 08 March 2023
- **Published in SciPost Phys. 15, 205 (2023):** 24 November 2023

3.13.5.8 Preparations for quantum simulations of quantum chromodynamics in <math display=”inline”><mrow><mn>1</mn><mo>+</mo><mn>1</mn></mrow></ dimensions. I. Axial gauge [95]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

3.13.5.9 Simulating Collider Physics on Quantum Computers Using Effective Field Theories [106]

- **Authors:** Christian W. Bauer, Marat Freytsis, Benjamin Nachman
- **Posted on arXiv:** 09 February 2021
- **Published in Phys.Rev.Lett.:** 18 November 2021

3.13.5.10 Quantum simulation of quantum field theory in the light-front formulation [107]

- **Authors:** Michael Kreshchuk, William M. Kirby, Gary Goldstein, Hugo Beauchemin, Peter J. Love
- **Posted on arXiv:** 10 February 2020
- **Published in Phys. Rev. A 105, 032418 (2022):** 08 March 2022

3.13.5.11 SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers [108]

- **Authors:** Natalie Klco, Jesse R. Stryker, Martin J. Savage
- **Posted on arXiv:** 19 August 2019
- **Published in Phys. Rev. D 101, 074512 (2020):** 22 April 2020

3.13.5.12 General Methods for Digital Quantum Simulation of Gauge Theories [109]

- **Authors:** Henry Lamm, Scott Lawrence, Yukari Yamauchi
- **Posted on arXiv:** 19 March 2019
- **Published in Phys. Rev. D 100, 034518 (2019):** 29 August 2019

3.13.5.13 Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers [110]

- **Authors:** Kubra Yeter-Aydeniz, Eugene F. Dumitrescu, Alex J. McCaskey, Ryan S. Bennink, Raphael C. Pooser, George Siopsis
- **Posted on arXiv:** 29 November 2018
- **Published in Phys. Rev. A 99, 032306 (2019):** 04 March 2019

3.13.5.14 Digitization of scalar fields for quantum computing [111]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 30 August 2018
- **Published in Phys. Rev. A 99, 052335 (2019):** 23 May 2019

3.13.5.15 Real-time dynamics of lattice gauge theories with a few-qubit quantum computer [112]

- **Authors:** E.A. Martinez, C.A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmonte, T. Monz, P. Zoller, R. Blatt
- **Posted on arXiv:** 15 May 2016
- **Published in Nature:** 22 June 2016

3.13.5.16 Towards Quantum Simulating QCD [113]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 25 September 2014
- **Published in Nucl.Phys.A:** 28 September 2014

3.13.5.17 Quantum Algorithms for Fermionic Quantum Field Theories [114]

- **Authors:** Stephen P. Jordan, Keith S. M. Lee, John Preskill
- **Posted on arXiv:** 28 April 2014

3.13.5.18 Quantum Computation of Scattering in Scalar Quantum Field Theories [115]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** December 2011

3.14 Quantum Storage

3.14.1 Track Reconstruction

3.14.1.1 Particle track classification using quantum associative memory [118]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

3.14.1.2 Quantum Associative Memory in HEP Track Pattern Recognition [125]

- **Authors:** Illya Shapoval, Paolo Calafiura
- **Posted on arXiv:** 29 January 2019
- **Published in EPJ Web of Conferences 214 (2019) 01012:** 2019

3.15 Quantum Unsupervised Clustering Algorithms

3.15.1 Anomaly Detection

3.15.1.1 Quantum anomaly detection in the latent space of proton collision events at the LHC [19]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

3.15.1.2 Unsupervised event classification with graphs on classical and photonic quantum computers [16]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

3.15.2 Jet Algorithms and Jet Tagging

3.15.2.1 Quantum clustering and jet reconstruction at the LHC [76]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

3.15.2.2 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [77]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

3.16 Quantum Walks

3.16.1 Event Generation

3.16.1.1 Collider events on a quantum computer [65]

- **Authors:** Gösta Gustafson, Stefan Prestel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 21 July 2022
- **Published in JHEP:** 07 November 2022

3.16.1.2 Quantum walk approach to simulating parton showers [66]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 28 September 2021
- **Published in Phys. Rev. D 106 (2022) 056002:** 01 September 2022

3.17 Tensor Networks

3.17.1 Jet Algorithms and Jet Tagging

3.17.1.1 Classical versus quantum: Comparing tensor-network-based quantum circuits on Large Hadron Collider data [78]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 21 February 2022
- **Published in Phys. Rev. A 106 (Dec, 2022) 062423:** 19 December 2022

3.17.1.2 Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States [74]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 15 June 2021
- **Published in JHEP 08 (2021) 112:** 23 August 2021

3.18 Variational Quantum Circuits

3.18.1 Anomaly Detection

3.18.1.1 Quantum anomaly detection for collider physics [21]

- **Authors:** Sulaiman Alvi, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 16 June 2022
- **Published in JHEP:** 22 February 2023

3.18.1.2 Anomaly detection in high-energy physics using a quantum autoencoder [18]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi
- **Posted on arXiv:** 09 December 2021
- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

3.18.2 Beyond the Standard Model

3.18.2.1 Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets [27]

- **Authors:** Miguel Caçador Peixoto, Nuno Filipe Castro, Miguel Crispim Romão, Maria Gabriela Jordão Oliveira, Inês Ochoa
- **Posted on arXiv:** 06 November 2022
- **Published in Front. Artif. Intell. 6 (2023) 1268852:** 20 November 2023

3.18.3 Detector Simulation

3.18.3.1 Precise image generation on current noisy quantum computing devices [33]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borras, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

3.18.4 Event Classification

3.18.4.1 Quantum Vision Transformers for Quark–Gluon Classification [53]

- **Authors:** Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva, Eyup B. Unlu
- **Posted on arXiv:** 16 May 2024
- **Published in Axioms 2024, 13(5), 323:** 13 May 2024

3.18.4.2 Quantum Metric Learning for New Physics Searches at the LHC [54]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

3.18.4.3 Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets [27]

- **Authors:** Miguel Caçador Peixoto, Nuno Filipe Castro, Miguel Crispim Romão, Maria Gabriela Jordão Oliveira, Inês Ochoa
- **Posted on arXiv:** 06 November 2022
- **Published in Front. Artif. Intell. 6 (2023) 1268852:** 20 November 2023

3.18.4.4 Higgs analysis with quantum classifiers [48]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

3.18.4.5 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [55]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

3.18.4.6 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [56]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

3.18.4.7 Event Classification with Quantum Machine Learning in High-Energy Physics [57]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

3.18.5 Event Generation

3.18.5.1 Partonic collinear structure by quantum computing [67]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

3.18.6 Jet Algorithms and Jet Tagging

3.18.6.1 Quantum Machine Learning for b-jet charge identification [79]

- **Authors:** Alessio Gianelle, Patrick Koppenburg, Donatella Lucchesi, Davide Nicotra, Eduardo Rodrigues, Lorenzo Sestini, Jacco de Vries, Davide Zuliani
- **Posted on arXiv:** 28 February 2022
- **Published in JHEP:** 01 August 2022

3.18.7 Neutrinos

3.18.7.1 Hybrid Quantum-Classical Graph Convolutional Network [92]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

3.18.7.2 Quantum convolutional neural networks for high energy physics data analysis [93]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

3.18.8 Quantum Field Theories

3.18.8.1 Quantum-classical simulation of quantum field theory by quantum circuit learning [116]

- **Authors:** Kazuki Ikeda
- **Posted on arXiv:** 27 November 2023

3.18.9 Track Reconstruction

3.18.9.1 Quantum-Annealing-Inspired Algorithms for Track Reconstruction at High-Energy Colliders [126]

- **Authors:** Hideki Okawa, Qing-Guo Zeng, Xian-Zhe Tao, Man-Hong Yung
- **Posted on arXiv:** 22 February 2024
- **Published in Comput.Softw.Big Sci.:** 28 August 2024

3.18.9.2 Charged particle reconstruction for future high energy colliders with Quantum Approximate Optimization Algorithm [127]

- **Authors:** Hideki Okawa
- **Posted on arXiv:** 16 October 2023

3.18.9.3 Particle track reconstruction with noisy intermediate-scale quantum computers [128]

- **Authors:** Tim Schwägerl, Cigdem Issever, Karl Jansen, Teng Jian Khoo, Stefan Kühn, Cenk Tüysüz, Hannsjörg Weber
- **Posted on arXiv:** 23 March 2023

3.18.9.4 Studying quantum algorithms for particle track reconstruction in the LUXE experiment [129]

- **Authors:** Lena Funcke, Tobias Hartung, Beate Heinemann, Karl Jansen, Annabel Kropf, Stefan Kühn, Federico Meloni, David Spataro, Cenk Tüysüz, Yee Chinn Yap
- **Posted on arXiv:** 14 February 2022
- **Published in J.Phys.Conf.Ser.:** 2023

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[16] A. Blance and M. Spannowsky, *Unsupervised event classification with graphs on classical and photonic quantum computers*, *JHEP* **21** (2020) 170 [2103.03897].

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